

PROBLEM STATEMENT

Design and develop surveillance drone capable of operating at 10K ft altitude with a maximum endurance of 48hrs.

Submitted by **Team INDIA**

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In partial fulfilment for Internship of
AEROSPACE VEHICLE DESIGN INTERNSHIP



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BONAFIDE CERTIFICATE

Certified that this project report “**Design and develop surveillance drone capable of operating at 10K ft altitude with a maximum endurance of 48hrs.**” is the bonafide work of “**Ravanam Hemanth Siva Kumar, L. Ram Kumar Narayanan, Pratap Kashinath Gujar, Akansha Raman**” who carried out the project work under my supervision. Certified further that to the best of my knowledge the work reported here does not form part of any other project/research work on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

Through this project, we intend to draft and design a **MALE UAV**.

This design project does not entail manufacturing UAV from the ground up. Rather, it requires adequate selection, customization, and modification of a pre-fabricated product.

The final design will be able to achieve sustained flight for a minimum of 36 hours with the added payload of GPS, altimeter, computer, autopilot, camera, etc. The body of the UAV is an RC plane.

Several designs are being considered. The final design choice will hinge on manufacturer specifications. Additionally, an adequate motor (liquid fuel vs. electric) will be chosen depending on the thrust-to-weight ratio required and the necessary range. The fuselage of the design needs to be sufficiently large enough to accommodate the electronic components necessary for autonomous flight. Wing loading is also of concern, thus the wing design needs to be appropriate.

Through this project, we intend to draft and design a **SURVEILLANCE DRONE** that can monitor **ENEMY BASES**.

A drone aircraft will be required to fly out of sight of the operator, but has zero intelligence, merely being launched into a pre-programmed mission on a pre-programmed course and a return to base.

Additionally, the final design of the SURVEILLANCE DRONE is presented together with some of its most important features like Transmission, Guidance, Navigation, Endurance, Power source selection, Propulsion selection and Manufacturing techniques amongst others.

I. Introduction

An unmanned aerial vehicle (UAV for short; also known as a drone) is any aircraft that does not have a human pilot onboard. UAVs have their origins as early as 1915 when Nikolai Tesla wrote a dissertation in which he described “an armed, pilotless-aircraft designed to defend the United States.” [1] UAVs come in a variety of sizes, designs and purposes. Initially, UAVs were merely remotely piloted; however, autonomous control is becoming more widely utilized.

The Surveillance Drone navigates using an internal [GPS navigation device](#), and either a pre-programmed flight profile (in which case the system is fully autonomous from take off to landing), manual override from a [ground control station](#), or a combination of both. It can autonomously return to base and land in case of lost communication with the ground station. The system has fully automatic launch and recovery (ALR) and all-weather capabilities.

The Surveillance Drone can carry an array of sensors, including [thermographic camera](#)(infrared) and visible-light [airborne ground surveillance](#), intelligence systems and various [radar](#) systems.

The Surveillance Drone is also capable of [target acquisition](#) and artillery adjustment.

The payload sensors communicate with the ground control station in real time, using either direct [line of sight data link](#), or via an airborne/satellite relay. Like the navigation system, the payload can also be used in either a fully pre-programmed autonomous mode, or manual real-time remote operation, or a combination of both.

- **Payloads**

- Electro Optical (TV & IR Combi or Triple Sensor TV/IR/LD)
- Synthetic Aperture Radar (SAR)
- Maritime Patrol Radar (MPR)
- COMINT & ESM capability
- Customer furnished sensor suites
- Communication relay package
- Integrated ATC Radio

Datalinks

- Direct Line-of-Sight (LOS) data link
- UAV airborne data relay for beyond LOS missions
- Ground-based data relay

Performance

- Endurance:48 hr
- Range:1000 km
- Range with satellite coverage:Beyond Line of Sight
- Altitude:12,000 ft

Technical Data

- Maximum take-off weight:1,150 kg
- Maximum Payload Weight:250 Kg
- Overall Length:8.50 m
- Wingspan:16.60 m

Motivation

Developing an unmanned aerial vehicle has been one of the main points of concern by many countries all over the world; about 70 different countries have some sort of UAV technology. UAV expenditures reached more than US\$ 58.5 billion and constituted a growth of more than 16% in 2021. Approximately 70% of global growth and market share is in the US. UAVs are used to gather information from the air in hostile areas. They can also be used in devastated areas where man support may not be available. These types of UAVs must be portable by ground and very reliable for recurrent use. With these types of uses by the military the UAVs designed are very costly and have very specific uses designed for each. The goal for the UAV design is to provide a cheaper alternative to these very costly military products. The UAV will be equipped HD camera with target recognition, a weight under 1750 kg., and a GPS guided autopilot to fly desired flight path.

Literature Survey

UAVs have been in production since before the Wright Brothers first took their historic flight. The earliest account can be traced back to the American Civil War, when an inventor patented an unmanned balloon that carried explosives that could be dropped after a time-delay fuse mechanism triggered a basket to overturn its contents [2]. While this is a relatively primitive idea of what the world has come to know today as “drones” it goes to show how early man began thinking about unmanned aerial systems. This technology began taking small leaps in the years following the American Civil War - the first military aerial reconnaissance photos were taken in 1898 during the Spanish-American War via a camera attached to a kite [2]. As the trend shows, many of the advancements in this technology arose during times of war, whether it was used to help with an offensive strike or just to acquire intelligence on enemy locations and activities. This is seen throughout the history and progress of unmanned aerial technologies. Advancements took place in Britain during the 1930s, where a radio-controlled UAV (dubbed the Queen Bee) served as aerial target practice for British pilots, and also during World War II, during which the Nazis developed an unmanned flying bomber known as the V-1 [2]. It wasn't until the 1970s that Israel

developed the Scout and the Pioneer, which started the development toward the more widely known glider-type UAVs [2]. It was from this design that the Predator drone came to be; the Predator is the most sophisticated UAV in existence to date, these drones have come a long way from the “balloons” of the past. It’s autonomous control networks show just how much this technology has evolved. The structural design of UAVs has changed over their developmental history in order to serve a variety of purposes. UAV design and advancement is a global activity. As technology and needs change, UAVs can be improved to serve these needs. There are several design considerations that are constant. The first of these design criteria is the degree of autonomy. Early UAV designs were mostly set to fly a specified path until they ran out of fuel. They carried a camera onboard, which would be recovered after the UAV landed. Later, the advent of radio control systems allowed UAVs to be piloted from the ground. Modern UAVs often combine these two basic 11 functionalities. These two modes of operation do not strictly signify autonomy. True autonomy suggests the ability of the aircraft to operate without human interaction. In this regard, UAVs are still very immature. UAV autonomy technology is divided into the following categories:

- Sensor fusion: On board the vehicle a combination of sensors are used.

- Communications: Communication and coordination will be handled between multiple sources in the existence of curtailed and imperfect information
- Motion planning (also called Path planning): Determining the optimal path for the vehicle in accordance with specific objectives and constraints such as obstacles go
- Trajectory Generation: Designed for optimal control and maneuverability to follow a particular route or to go from one place to another.
- Task Allocation and Scheduling: Set the optimal distribution of tasks between a group of agents, with time constraints and equipment limitations.
- Cooperative Tactics: The optimal sequence and spatial distribution of activities between agents in order to make the most of the chances of success in any case or situation.

[3] The ultimate goal of UAVs is to replace human pilots altogether.

Another major design criterion is UAV endurance (range). Since there is no human pilot onboard, there is no concern for pilot fatigue. UAVs can be designed to maximize flight times to take advantage of this fact. Different systems can afford a wide variety of maximum range. Internal combustion engines require relatively frequent refueling and inflight refueling is a major obstacle for this type of propulsion system. Photovoltaic UAVs offer the potential for unlimited range and there is much research in this field. One more type of fuel system is hydrogen, which is proposed for use with certain models of stratospheric persistent UAVs. The

AeroVironment's Global Observer is one such UAV. This aircraft runs on hydrogen and has a range of 7 days. The idea is for two of such UAVs to be used in tandem to provide continuous, uninterrupted operation 365 days a year.

With the sophistication that these systems have arrived at, the market for them has grown astronomically. While the United States still has the largest stockpile of unmanned aircraft, the rest of the world is beginning to follow suit. More than 50 countries have purchased surveillance drones, and many have started in-country development programs for armed versions [4]. More than two-dozen different models were shown at a recent aviation show in China [4]. Due to the changing landscape of the theater of war, many nations are leaning toward unmanned aircraft to handle delicate situations in which human lives need not be put at risk. Also, taking into account the fact that drones sell for a fraction of the cost of manned airplanes, the amount of UAVs a nation can purchase at once has enticed many nations into entering the drone zone.

In general, UAVs fall into one of six functional categories:

- Target and Decoy: simulating enemy missiles or aircraft for ground and air gunnery
- Reconnaissance: battlefield intelligence gathering
- Logistics: cargo and logistics application

- Research and Development: used for UAV technology development 14
- Civil and Commercial: specifically designed for civil and commercial applications.

The basis of the competition is a reconnaissance mission for the INDIAN Troops for specified target accusation as well as added fight directions mid-flight. The story is that an island has had storms and pirates have invaded the island. The UAV is to stay within a specified area and transmit the images back to base as well as locations for desired targets.

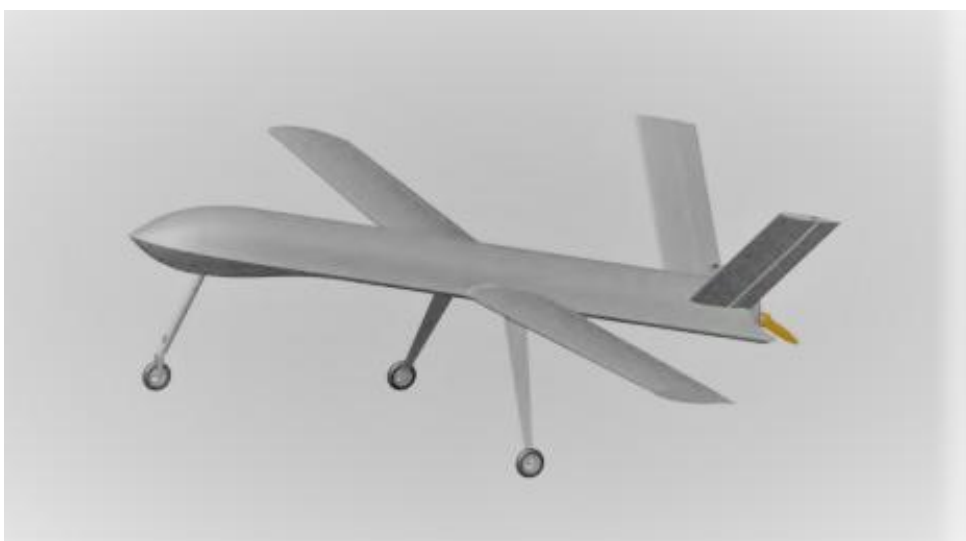
II. Design

Overview

In order to decide upon a design for this UAV, certain limiting factors had to be taken into consideration, namely the payload of electronic equipment that would need to be housed and secured somewhere within the fuselage. Also by taking into account the necessary minimum flight time this plane would need to see, while carrying this load, the possibilities were narrowed down further. After conducting a little research into the designs of commonly used UAVs and comparing their uses with those that this plane would see, it was decided that a conventional glider design would be used. Being that this plane would need to see at least twenty minutes of flight time the glider seemed like the most feasible option. In the design of a glider long periods of flight can be sustained due to the aircraft's lift to weight ratio; as the wings of a glider are lengthened it becomes capable of bearing more weight during flight. Since gliders fly mostly at low speeds, choosing a glider also indirectly helped to alleviate some of the confusion in choosing between electric and gas powered engines. Due to the fact that gliders come in a variety of configurations, below are a few conceptual designs that were weighed out before deciding upon the final design.

Design:

The seen design below in figure has become the most commonly employed design for UAV technology today. This design consists of a glider-type fuselage that has a round nose that is lofted into a thin, circular shape that is met by a V-tail used for increased maneuverability. Similar long, thin, rigid wings are seen in this design as were seen in the glider. The difference between this design and that of the glider comes from its extended fuselage, which allows for a back centered propeller to be used. This is a significant improvement from the other designs, because it allows for the camera to be placed in the nose of the plane, as well as all of the other electrical components, without being interfered with by the propeller.



Proposed Design

After looking into many possible options and comparing the pros and cons between overall weight, flight time, it was decided that the design was the best design to go with. With its rigid frame, light weight, and adherence to design criterion this plane was the best possible choice to use for our flight conditions.

Last thing that needs to be decided upon is the type of engine that will be used to fly it. There are both positive and negative aspects related to an electrical motor and a gasoline powered motor, the greatest factor rests in the weight that would be added to the system. Once an estimate on the weight of all of the internal components for the on-board autopilot, batteries and processing equipment is received a decision will be made on the type of engine to be used for this design.

III. Analytical Analysis

Understanding problem

At this step the team task was to identify all the distinctive features of the structure and the principles of the design objectives. The type of loading inherent in the problem and the importance of any other environmental influences are also taken into account.

Intended User(s) And Intended Use(s)

- Intended user(s).

The primary users of the unmanned aerial vehicle (UAV) are mostly military, but not limited to the work of civilian personnel. Many researchers use UAV to survey locations for mapping purposes, pipeline security, and homeland security. The use of UAV in commercial aerial surveillance is expanding rapidly. Police department can use the UAV if an IR/Heat sensor is attached to find “Grow Houses”. Other possible users are teams of investigators that need to enter environment that might be contaminated with harmful toxins, because a UAV is unmanned it eliminates some chance of contamination.

Intended use(s).

The team will be creating an UAV that will follow a path and it will seek and identify colorful and alphanumeric targets, possible by using camera and sensors. Other uses can be focus on the aid of ARMY department, using a UAV that has alphanumeric capability it can find targets that can range from a simple license plate to even a missing person. Since the UAV can be given a path to follow and it is also quite when compared to helicopter, it can also be used as an eye in the sky for multiple purposes, such as, surveillance of a woodland area where noise can have negative affects. Unmanned Aerial Vehicles can be used to enter contaminated environment, and can be used to scan the area. With the right sensors attached to the UAV, a UAV can scan for levels of contaminates. An excellent example of a possible use of a UAV could have been in Japan after the earthquake. A UAV could have aided in the search of human life. Also, it could have been used to know the level of radiation in areas of interest that came from the damages of power plants with the proper sensors.

Mathematical Model

Once the team determined the essential features of the physical problem have been identified it was necessary to translate these features into a mathematical representation of the problem. For our problem this two stages involved two major tasks firstly, definition of the problem domain and secondly, selection of a mathematical formulation which best represented our physical behavior of the structure. The main equations to look at are those of lift and drag. In the two Equation 1: Drag and Lift Force Equation equations, A is representative of the surface area generally associated with the wings of the aircraft. The terms C_D and C_L are the coefficients of drag and lift, respectively. These coefficients are dimensionless numbers whose values have become almost standard through years of testing. In the free body diagram below, the various forces acting on an aircraft at any given time can be seen.

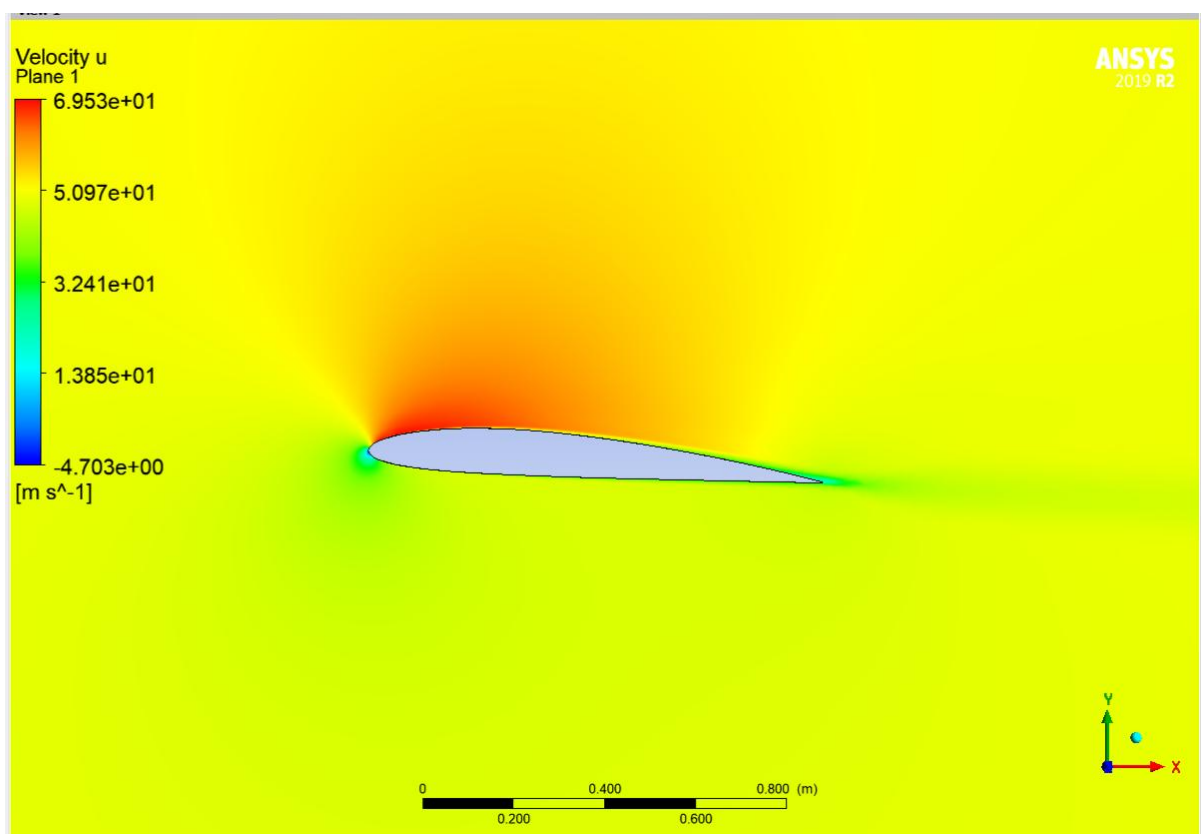
Equation 1: Drag and Lift Force Equation

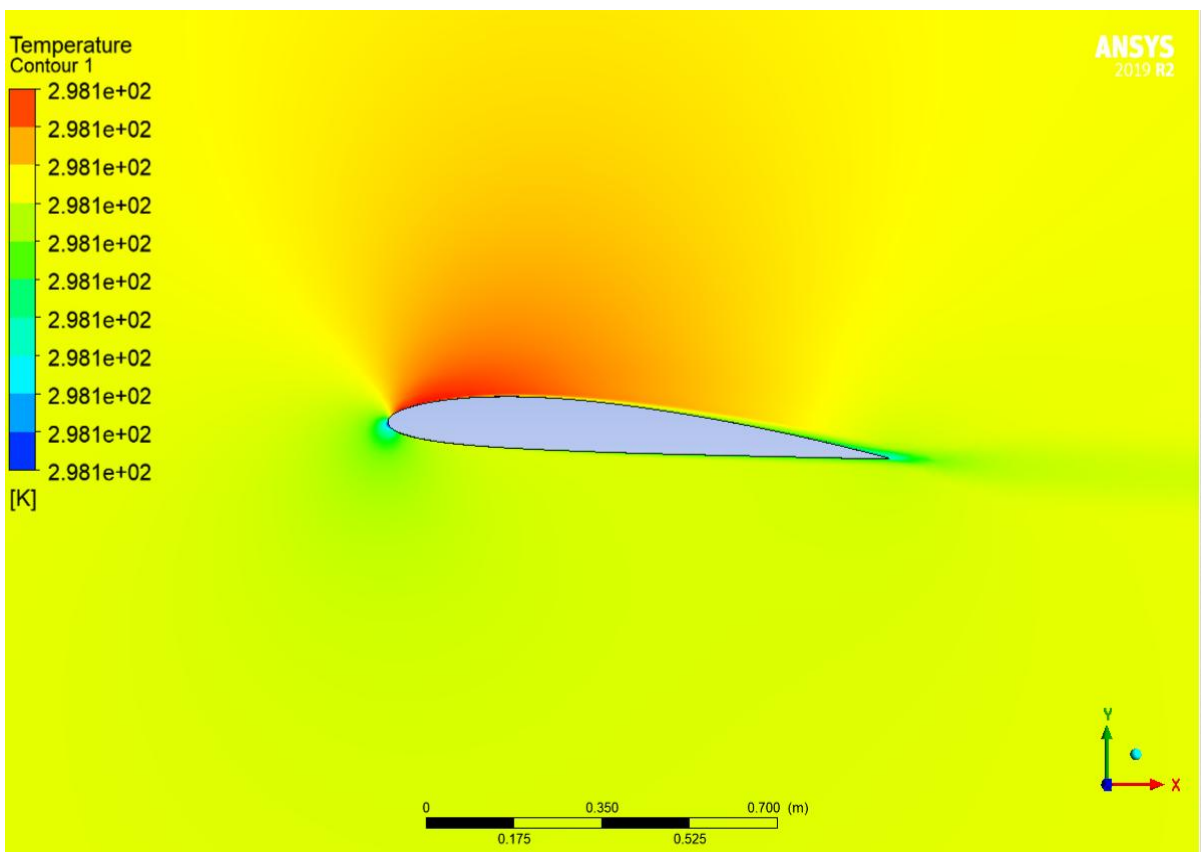
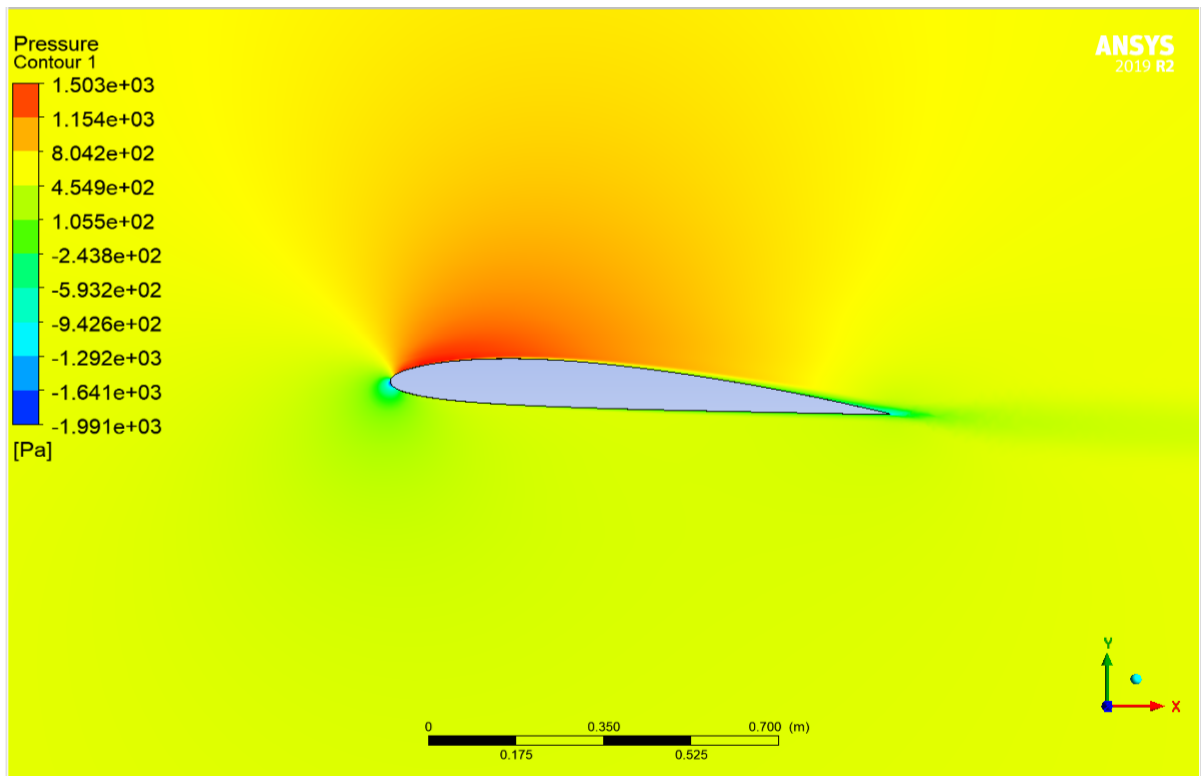
$$F_D = \frac{1}{2} \rho v^2 C_D A \qquad L = \frac{1}{2} \rho v^2 A C_L$$

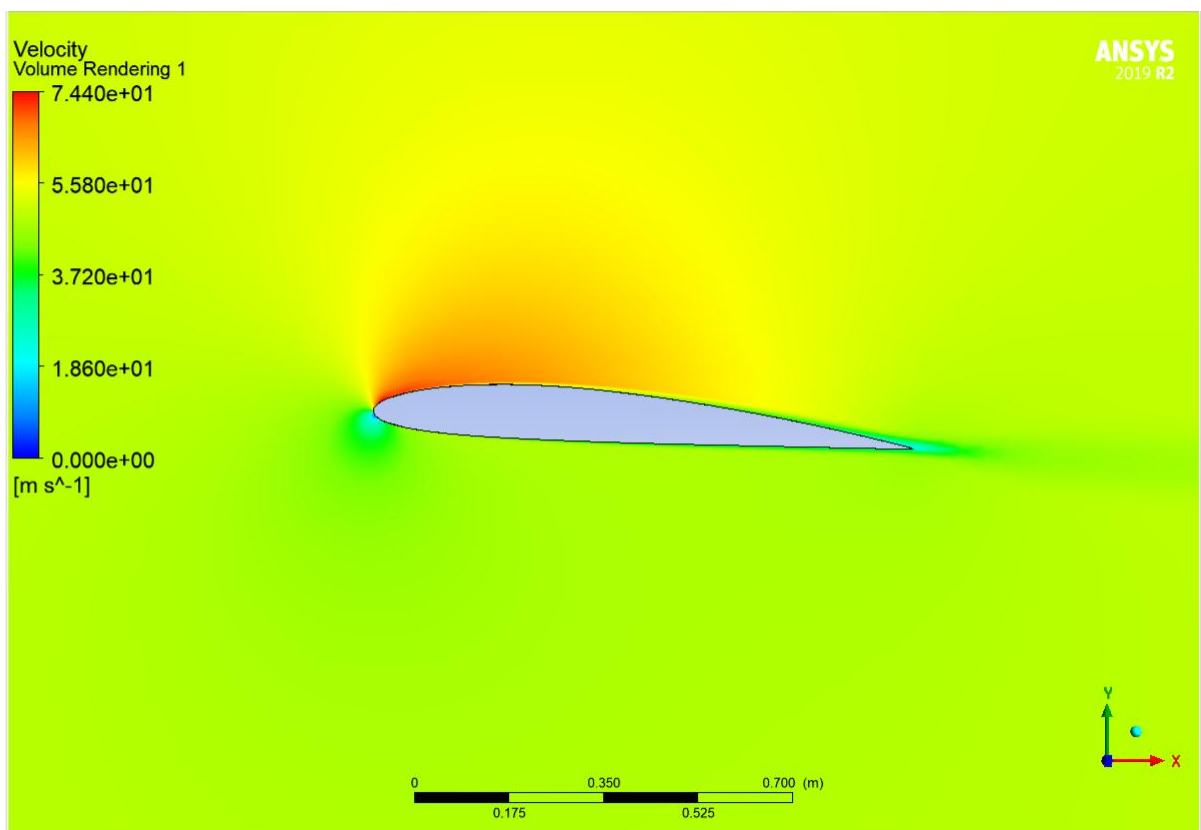
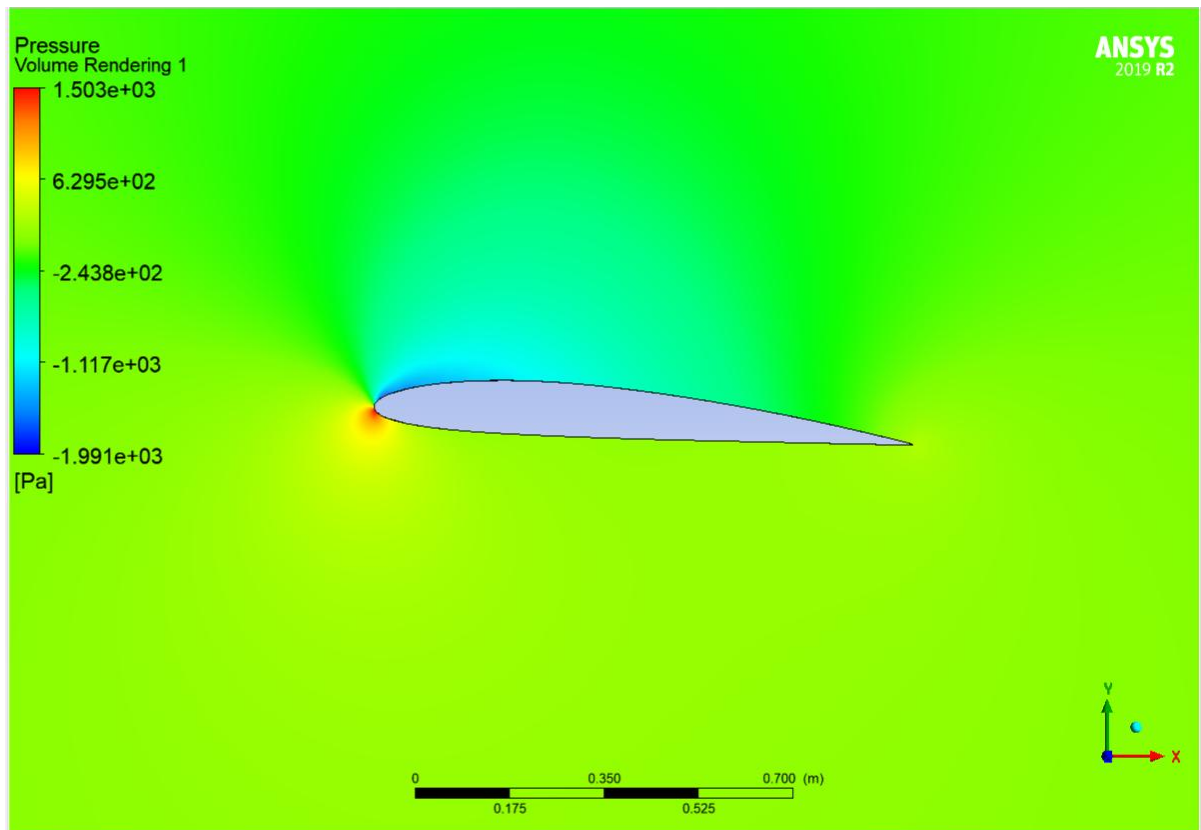
It can be seen that in order for an aircraft to achieve flight it must first overcome the forces of drag and its own weight via thrust and the lift associated with the design of the airfoil. The thrust of these planes is dependent upon the engines used to power them. Thus,

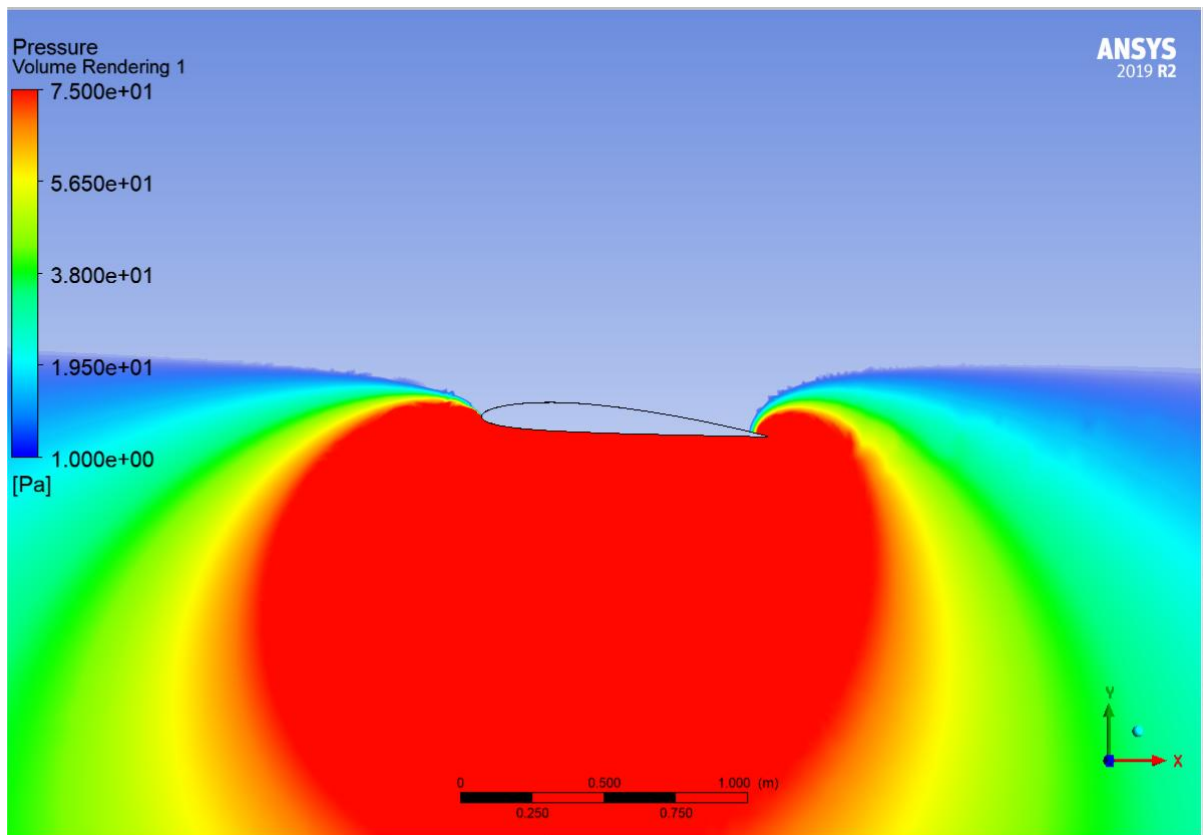
when designing a plane, the right geometry and size must be chosen for the fuselage and the airfoils, in order to calculate the drag and lift that would be associated with them; once the drag, lift, and weight are known the proper size engine can be chosen to power the plane.

To ensure the validity of this design before implementing it as our desired end product, a finite element analysis was conducted in SolidWorks. For this analysis the fluid flow over the airfoils, as well as the fluid flow over the fuselage, were tested to observe the aerodynamic properties of the aircraft.







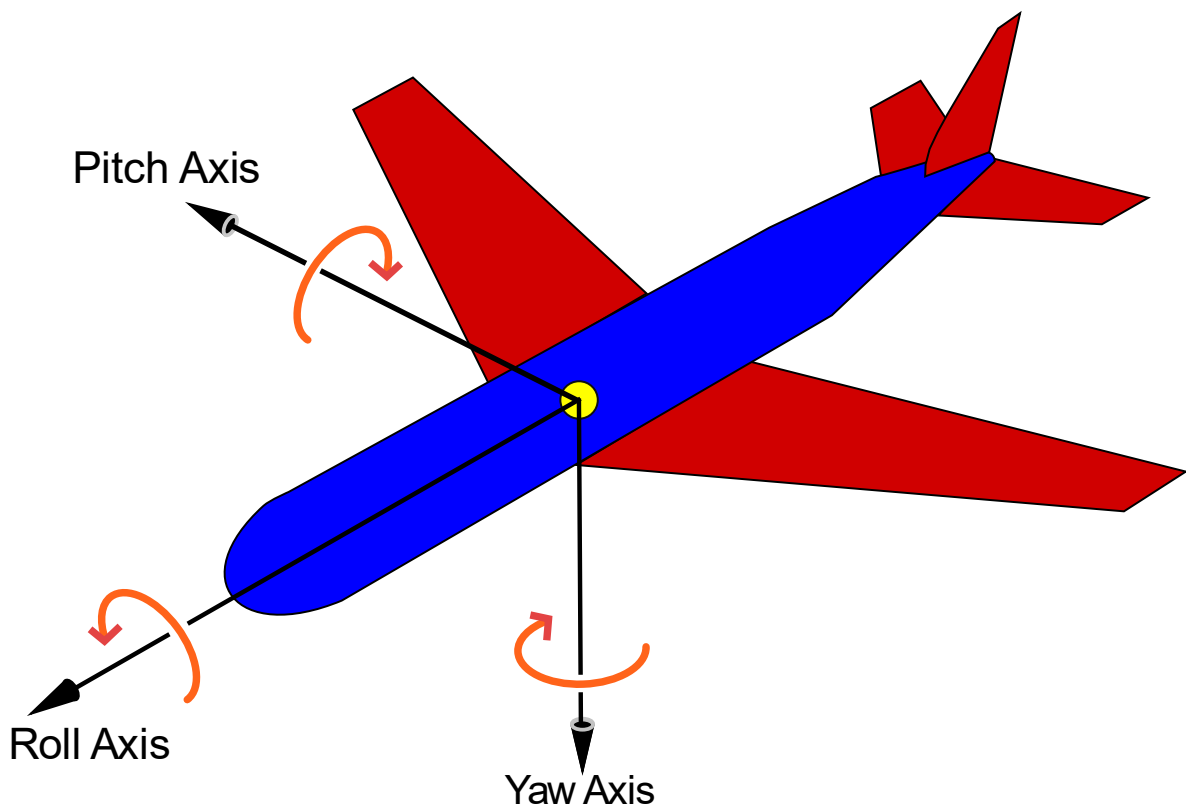


We have done this simulation pictures in Ansys 2019 R2, using CFD Post .

We have used NACA 2410 as aerofoil for our Design.

IV. Major Components

There are many important aspects to put into perspective when designing this aircraft the main components to take into account, however, are the fuselage, the wings, the stabilizers and the motors (both servos and driving). Each of these components has a contributing factor in some form of aerial rotation, also known as yaw, pitch, and roll.

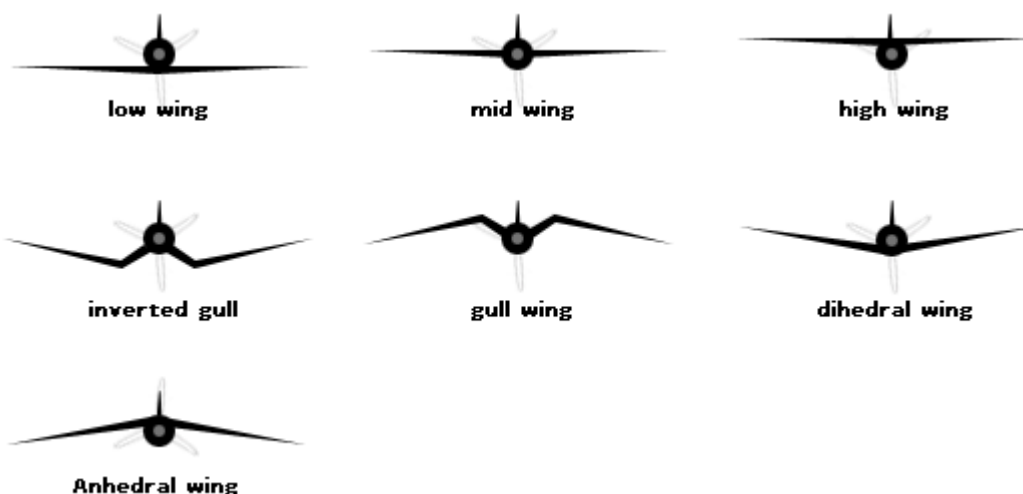


The pitch of an airplane is the nose up or down action of the airplane and is taken in relation to the axis running from wing to wing. The yaw component of rotation describes the motion of the nose of a plane from either left or right in relation to a vertical axis running

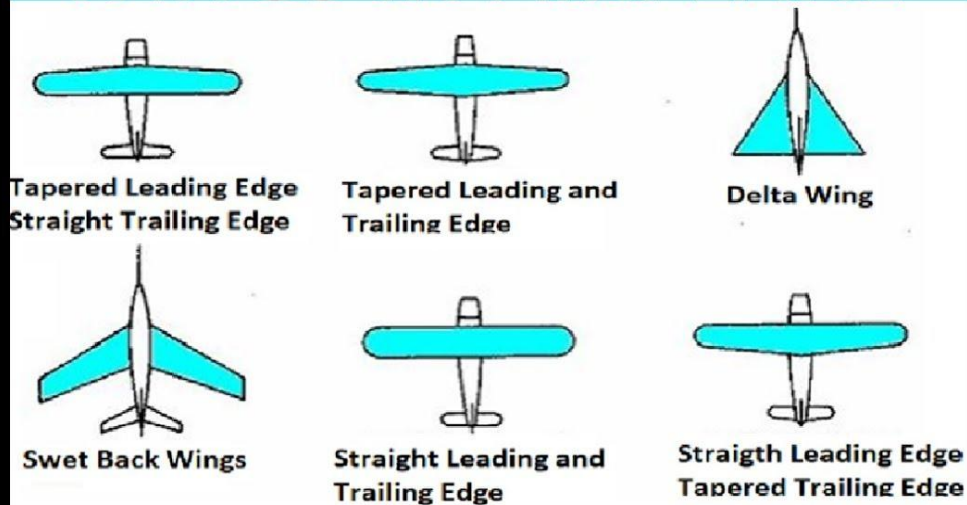
through the aircraft. Measured in relation to an axis running from nose to tail, roll is the rotation of a plane about this axis. By examining each of these components individually a conceptual design may be achieved.

Wings

The wings of a plane are necessary to provide lift and essentially allow the plane the ability to fly. In addition to creating the lift force that allows a plane to fly, the wings also control the roll of an airplane through the ailerons attached to their trailing edges. There are many different wing configurations and geometries that are used today in common airplane designs; these possibilities are displayed in the figures below.



DIFFERENT TYPE OF WINGS IN AIRPLANE



The main thing to take into account when choosing the right wing configuration is the function that the plane will be tasked to perform. Sweptback wings and delta wings are generally used for high speed flight and do not perform as well as straight wings in the regime of low, cruising speeds. Wing geometry is also largely based on the type of propulsion system that is used to power the aircraft because the thrust provided by this system provides a good idea of the planes maximum speed capabilities. For these reason a mid-plane, straight wing design was used.

- **Straight Wing:**

Contains features desirable for UAV based on low speed performance and stability during flight. This configuration has no forward or backward inclination that makes it optimal for our project.

Stabilizers

Along with contributing to the overall lift gained during flight, the stabilizers of an aircraft help to maintain a straight and level path, facing into the direction of the wind, throughout the course of flight. There are multiple designs that may be implemented and their functions will differ based upon the geometry of their configuration.

Fuselage

The main functions of an airplanes fuselage are to carry the crafts payload and to hold all the pieces of the plane together. Just like the wings, the design of the fuselage is dependent on the duties that the plane will be commissioned to perform. For a plane being designed to reach high rates of speed a slender, streamlined fuselage will be used to reduce drag associated with high-speed flight [8]. A plane being built to operate at low speeds while transporting a payload may be long and hollow, the shape all depends on the drag coefficient that comes along with it.

Fuselage

List of components of Surveillance Drone:

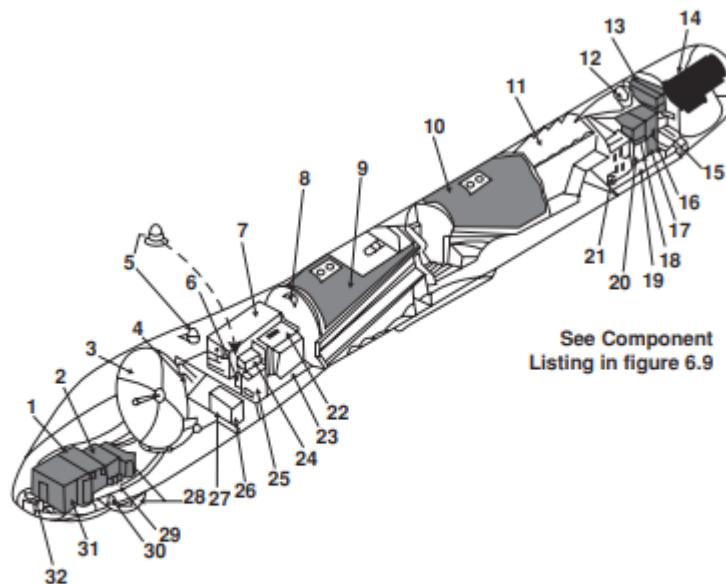


Figure 6.8 Internal view of MALE UAV and listing of components

1. Synthetic Aperture Radar (SAR) Antenna
2. Inertial Navigation System/GPS
3. Ku-Band Satellite Communications Antenna
4. Video Cassette Recorder
5. GPS Antennas (Left and Right)
6. Identification Friend or Foe Transponder
7. Ku-Band Satellite Communications Sensor Processor Modem Assembly
8. C-Band Upper Omni-directional Antenna Bracket
9. Forward Fuel Cell Assembly
10. Aft Fuel Cell Assembly
11. Ancillaries Bay
12. Ancillaries Cooling Fan
13. Oil Cooler/Radiator
14. Engine
15. Tail Servo (Left and Right)
16. Battery Assembly
17. Power Supply
18. Battery Assembly
19. Aft Equipment Bay Tray
20. Secondary Control Module
21. Synthetic Aperture Radar Processor / Electronics Assembly
22. Primary Control Module
23. Front Bay Avionics Tray
24. Receiver/Transmitter
25. Flight Sensor Unit
26. Video Encoder
27. De-ice Controller
28. Electro-Optical/Infrared Sensor Turret / Electronics Assembly
29. Front Bay Payload Tray
30. Ice Detector
31. Synthetic Aperture Radar (SAR) Receiver/Transmitter
32. Nose Camera Assembly

Figure 6.9 Internal view of MALE UAV and listing of components

ENGINE STUDY:

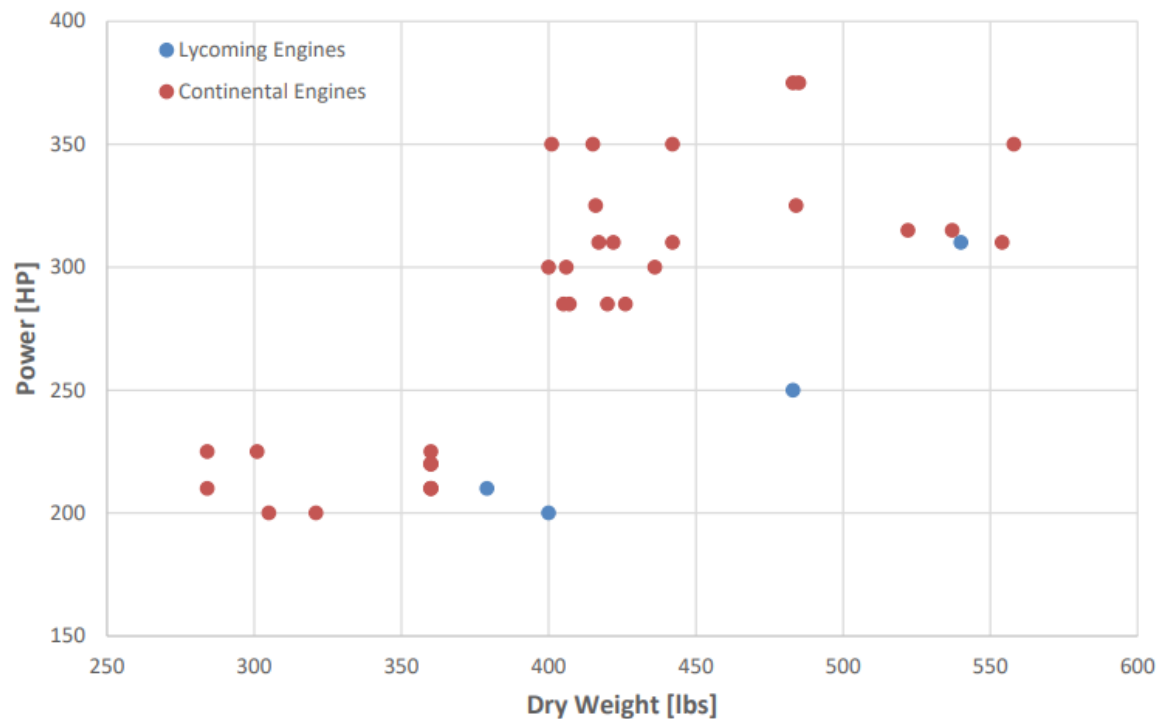


Figure 6: Weight as a function of power for turbocharged engines³

Visual detection of MALE UAV:

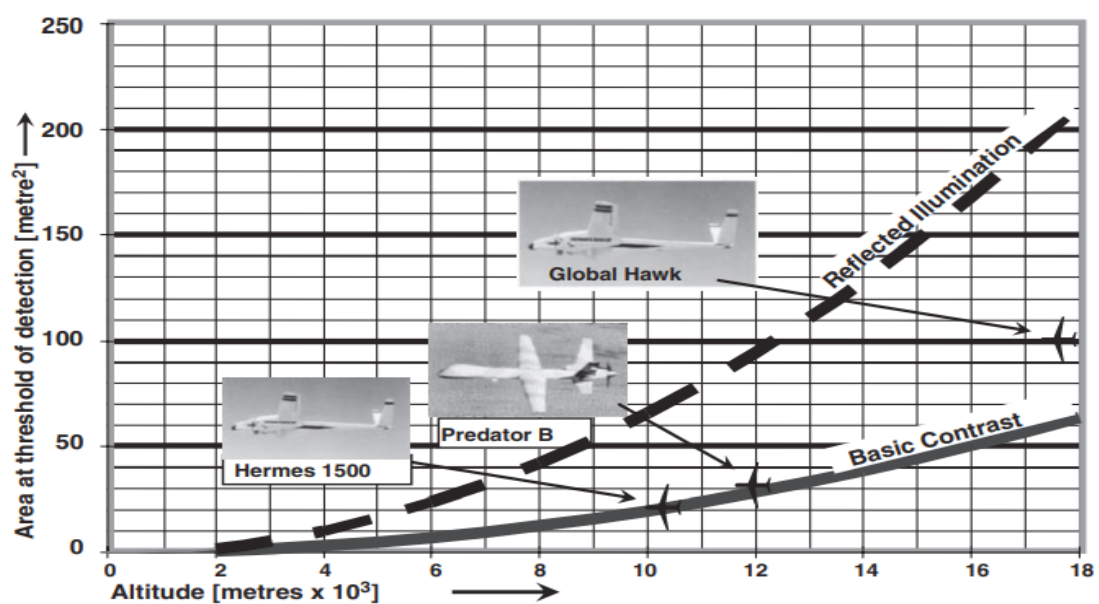


Figure 7.1 Threshold of visual detection (high altitudes)

Non dispensable Payloads

The approach taken here is to describe the various payloads whose design will be driven by several factors which will include:

- 1.the range, endurance and altitude required in the operation of the platform
- 2.mass and drag constraints imposed by a selected platform performance
- 3.installation and field-of-view constraints
- 4.type of target or surveillance modes
- 5.range and area of surveillance needed
- 6.resolution of the imagery
- 7.discrimination of the target from the background
- 8.need for tracking (track while scan, lock-on, etc.)
- 9.need for map data base correlation
- 10.need for recording
- 11.need for superimposed latitude/longitude, date, time data for admissible evidence

Electro-optic Payload Systems

These range from simple monochrome or colour single-frame cameras through colour TV systems, lowlight-level television (LLLTV), (also sometimes abbreviated to LLTV), thermal imaging video systems to multi-spectral systems.

Daylight Cameras

The light is focused through ground-glass lenses onto receptors (charged-coupled devices, CCD) from which the image is transformed into a format which can be displayed onto a monitor or coded on to a radio carrier frequency for transmission to a receiving station.

Low-light-level Cameras

LLL cameras function in the same manner as standard optical cameras, but are fed an amplified level of light, using fibre optics which accept light from a larger area and concentrate it onto the camera lens.

LLTV cameras can provide images at as much as one-tenth of the light necessary for the operation of ordinary cameras.

Thermal Imagers:

For thermal imaging cameras, only the two divisions of about 3–5 μm and about 7–15 μm are significant. These are usually known as short-wave IR and long-wave IR respectively. This is because absorption of infrared emission by water molecules in the atmosphere is at a minimum at these wavelengths, which provide a ‘window’ through which heat can be transmitted most effectively.

Resolution

The area of the receptor of the camera and the number of picture elements, pixels, that it contains will determine the acuity or resolution of the image. The larger the receptor area, the more light (or heat) it will accept and so will operate successfully at lower levels of target illumination. The more pixels receiving light from a target area so will the sharpness of the image be improved.

Johnson criteria which attempt to define the number of pixels needed in two coordinates:

Detection – something is there,

Recognition – (for example) it is a vehicle,

Identification – it is a Humvee, Landrover, Centurion tank, etc...

Lenses

Lenses for visual light cameras are ground from high-quality optical glass and the majority of cameras employ zoom lenses

Electro-optic Systems Integration

Mounting Electro-optic systems may be mounted in one or more of three ways.

- a) Essentially forward-looking from a mounting in the nose of the aircraft for purposes of obstacle avoidance.
- b) A rotatable turret is mounted beneath the aircraft to obtain a 360° azimuth field of regard for reconnaissance/surveillance.

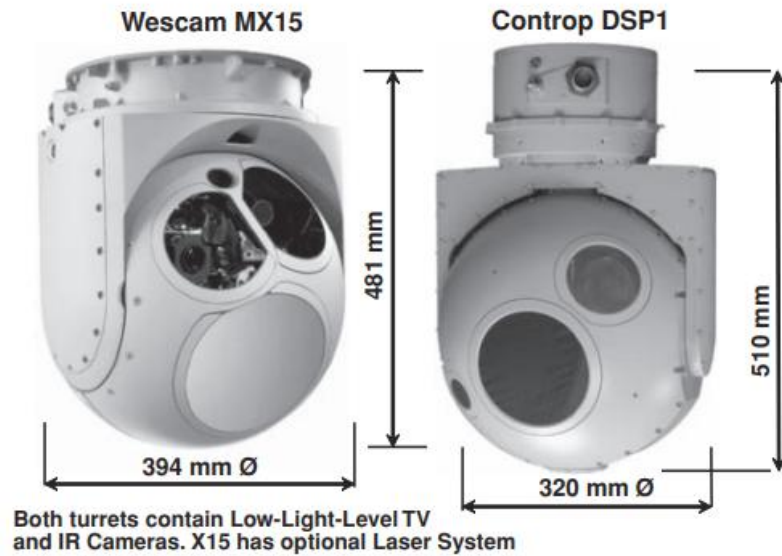


Figure 8.4 EO turrets for HALE and MALE UAV. (Reproduced by permission of Controp Precision Technologies Inc.) *Source:* Wescam

STABILITY AND CONTROL

- a) direction,
- b) horizontal speed
- c) altitude,
- d) rate of climb

The direction of flight (or heading) will be controlled by a combination of deflection of the rudder(s) and ailerons.

The horizontal speed will be controlled by adjustment to the propulsor thrust and elevator deflection.

The rate of climb to a given altitude is achieved by the application of a combination of elevator deflection and propulsor thrust

Design the aircraft to be aerodynamically neutrally stable with, in particular, little or no rotation generated by the fixed aerodynamic surfaces in response to gusts. The response now becomes one mainly of translation, so reducing the angular stabilisation requirements for the sensors. The movable control surfaces are used to steer and stabilise the aircraft in the normal manner relative to spatial coordinates.

ADVANTAGES

1. providing a steadier platform for payload functions
2. neutrally aero-stable design is that the aerodynamic tailplane and fin surfaces

NAVIGATION:

popular method of position fixing and navigation between points is by use of the Global Positioning System (GPS).

INDIAN IRNSS SYSTEMS is chosen for our Surveillance drone.

Costs:

(A) FIRST COSTS

The UAV equipped with surveillance sensors can be typically only 3–4% of the weight, require only 2.5% of the engine power (and 3% fuel consumption) and 25% of the size (wing/rotor span) of the light aircraft

The overall result is not obvious but, depending upon the surveillance requirements, may be of the order of: UAV 20–40% of manned aircraft cost UAV control station 20–40% of manned aircraft cost UAV + UAV control station 40–80% of manned aircraft cost.

(B) OPERATING COSTS:

Approximate UAV cost as % of manned aircraft cost:

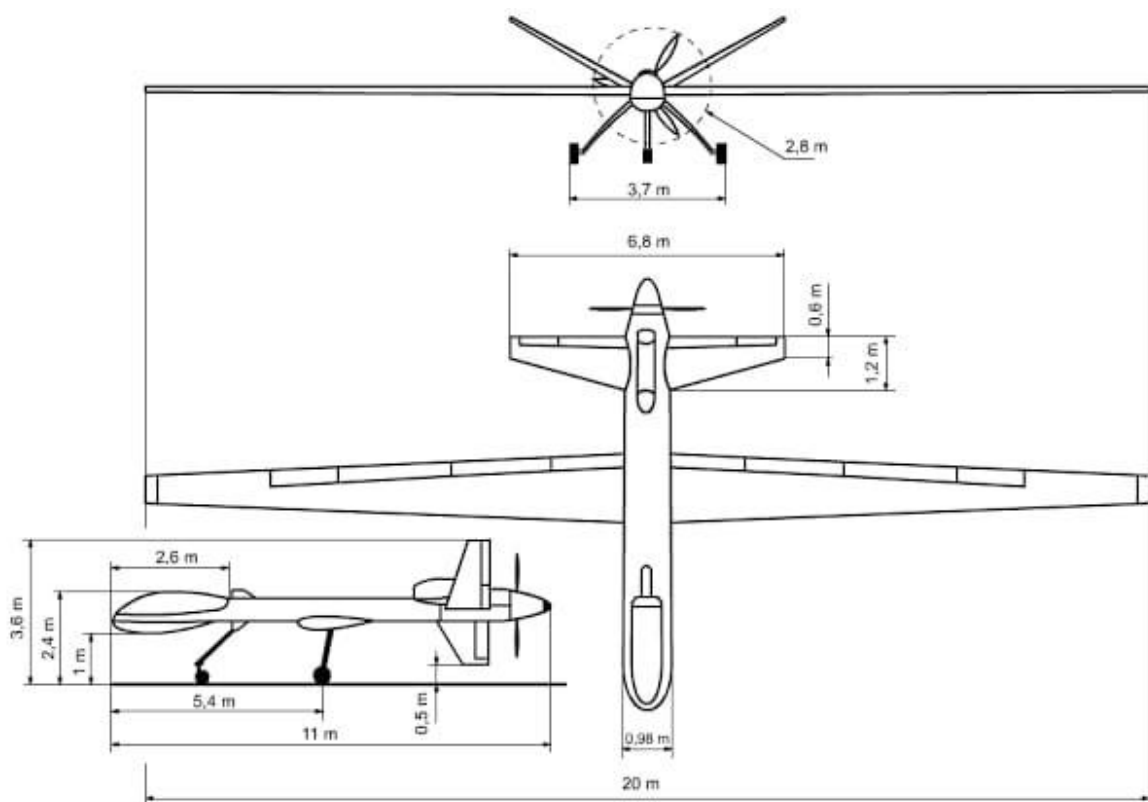
i) Interest on capital employed	40–80%	] Probably about 40% or less overall
ii) Depreciation	30–60%	
iii) Hangarage [includes support vehicle and CS]	20%	
iv) Crew salaries and associated costs*	50%	
v) Fuel costs	5%	
vi) Maintenance	20%	
vii) Insurance**	30%	

V. Structural Design

When designing a part it is important to reflect on the idea on how shape could be used to modify the ways in which material behave. Furthermore, a design must comply with key principle known as the F3 or “form, fit and function”, which is based on the principle that if the specifications, or criteria, for form, fit and function of a particular item are met, all other attributes, from an engineering design process, are moot or extraneous. In this section we will focus our attention on how the form or shape of the material can be used effectively in order to achieve a design goal. By “shaped” we mean that the cross section carry bending, torsional, and axial-compressive loads more efficiently than solid sections. The idea of shape is very important to our project as a result that all our designs have weight limitations. Therefore, it is essential that every design made can carry a given load condition by using the as little material as possible. An example of this idea of reducing weight or material by using “simple shapes” such tubes, I-sections and hybrid panel is visible on most of the component design for the completion of this project. As a result of that we have T-beam cross section for the main structural frame, carbon fiber honeycomb structures for floor and walls of the POD, titanium brackets for a higher structural integrity. In addition to the shape element of any design process, there is

another important aspect to the structural design, which is related to the principle of manufacturability. This aspect of the design process that has to do with the manufacturing process is very important when design something which will be explain in great detail on the subsequent section of this report.

DIMENSIONS



Desired Dimensions and Performance Characteristics

Length	: 9 m
Wing span	: 17 m
Max take-off weight	: 1750 kg
Max speed	: 250 kmph
Service Ceiling	: 9500 m
Endurance	: 36 hrs
Operational altitude	: 27500 ft

To get the above values we have taken 5 MALE UAV's and compared their specifications and taken average of their specification values

UAV's we have considered :

1. IAI Heron
2. Elbit Hermes 900
3. Northrop Grumman firebird
4. MQ-9 Reaper
5. MQ-1 Predator

Prototype System Description

As an unmanned aircraft, this system will contain many electrical and computer components used to stabilize and control its flight. This system will be subjected to user-controlled takeoff and landing, but during flight it will maintain complete autonomous control. An autopilot will be housed in the core of the craft controlling all of the servomotors and reacting to data being fed to its processor via the camera, as well as changing wind speeds and directions. This system will contain a series of relays, which keep a line of contact with mission control on the ground as seen in the figure below.

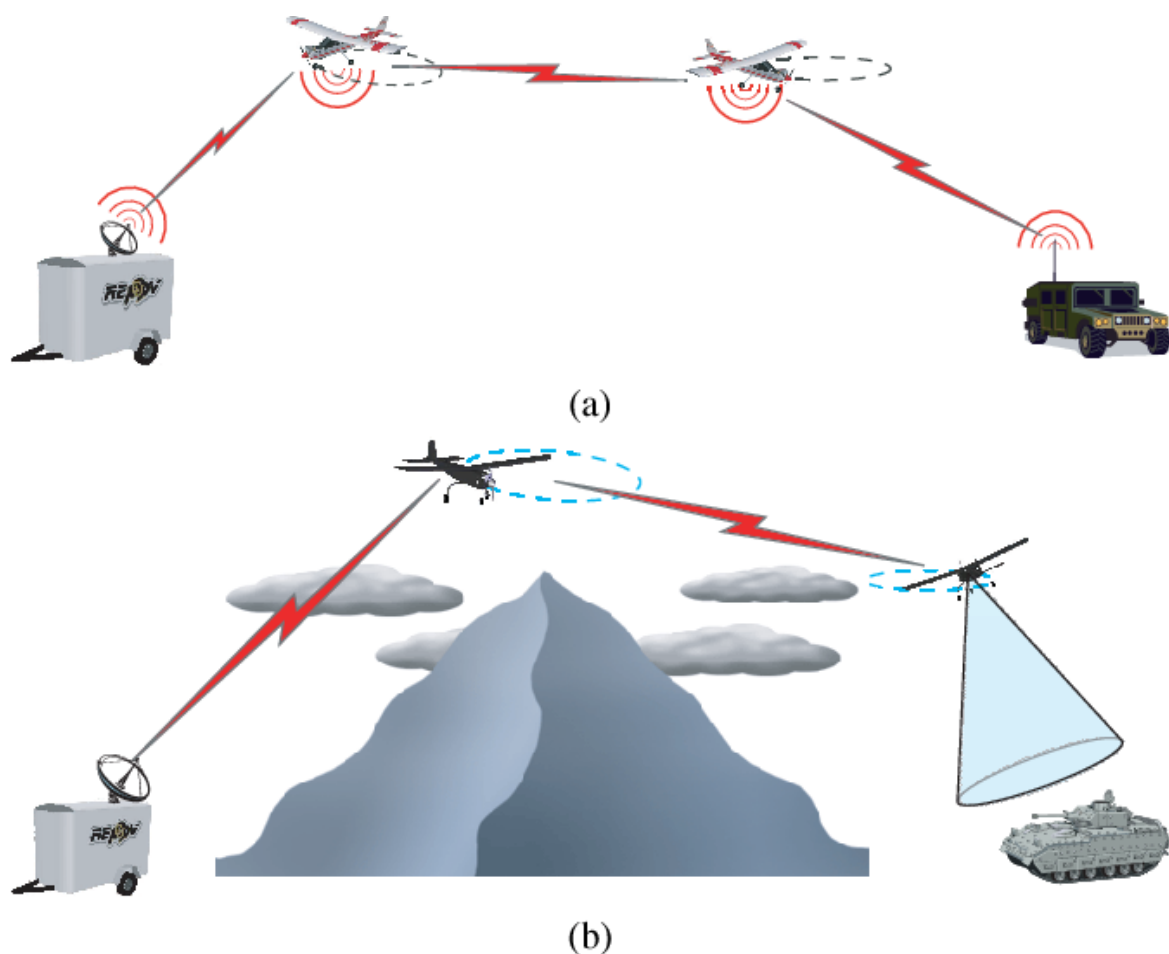


FIG. 1 Application of UAVs in communication relay chains (a) Extending UAV Computer Relays Also within this system will be a series of

sensors to help stabilize plane within its programming interface, these sensors include gyroscopes, pressure gauges, Pitot tubes and an image recognizing camera. These sensors will be mirrored on the programming interface so that the flight can be monitored at all times from an auxiliary location; a safety cut-off will allow the monitor to remotely operate the plane if anything seems out of the ordinary. These sensors will provide a constant feedback to the operators on the ground to ensure that all of the systems are operating correctly and to make sure that the target images are being accurately collected to abide by the rules of the competition.

- In regards to global culture, the military role of Unmanned Aerial Vehicles is growing at unprecedented rates. The rapid advances in technology are enabling the capability to be placed on smaller air frames which in turn increases the number of unmanned aerial vehicles and systems being deployed and used on the battlefield worldwide. Furthermore, as the capabilities grow for all types of Unmanned Aerial Vehicles, nations continue to subsidize their research and deployment leading to further advances, which enables them to perform a multitude of missions.

Their roles have expanded to assist in different missions such as:

- Target and decoy – providing ground and aerial targets to simulate an enemy aircraft or missile

- Reconnaissance – providing battlefield intelligence
- Combat – providing attack capability for high-risk missions
- Research and development – used to further develop UAV technologies to be integrated into field deployed UAV systems
- Civil and Commercial UAVs – UAVs specifically designed for civil and commercial applications. When successful, our project will further contribute to the implementation of innovative UAV systems to assist in being used for the different roles above.

VII . Global Learning

As UAV technology becomes more prominent in the global market, many areas of concern will begin to arise, some of which are already under scrutiny in the public eye. While the implementation of these craft in the military sector has led to a lower occurrence of human casualties, the question will still remain as to how these covert strikes will change the rules of engagement. Many of the targeted attacks being carried out today, in this countries war on terror, are being touted as assassinations; there has to be a strong case for justification, if not what would stop other countries from committing the same acts? While the United States was once known to have a monopoly in the military robotics industry, that time has come to an end. It is cited that between 75 and 87 countries have used unmanned aircraft in their militaries. Facts like these make it increasingly important to understand the repercussions that systems such as these can have on the world. It has been stated by some remote operators of such aircraft, that the use of this sort of technology has removed, completely, any sense of humanity associated with death while conducting a strike on a target. If combat at this level has become as surreal as these operators suggest, what will happen to the field of war? It is imperative that a series of rules be put into place on an international level to regulate

the use of these vehicles during combat. Another concern facing the industry of unmanned aerial vehicles is the level of autonomy available within them. As these systems move closer and closer towards complete autonomy how will their duties become affected?

In 1942 Isaac Asimov, a science fiction author, fashioned a list of laws regulating robotic behavior.

These “Three Laws of Robotics” exist in the following order:

1. A robot may not injure a human being or, through inaction, allow a human being to come to harm.
2. A robot must obey the orders given to it by human beings, except where such orders would conflict with the First Law.
3. A robot must protect its own existence as long as such protection does not conflict with the First or Second Laws.

While these laws were created merely as a theme for a science fiction novel, the message they present still remains apparent – at what point will it become acceptable procedure for an autonomous robot to decide the value of a human's life? And if the algorithms that run these systems become more and more sophisticated, will we eventually lose control over them? This idea has been portrayed in movies such as “I, Robot” and “Eagle Eye” merely to show that as the circuitry and programming developed for these robots becomes more sophisticated an artificial intelligent may in turn become prevalent. While this topic may be years ahead of its need for examination the threat that it may pose could be great. It is our duty

to make sure that the gap between human interaction and autonomy never becomes too big that we may lose control.

VIII . Conclusion

For the purposes of this project, a mid-winged airplane with significant amount of wingspan in comparison to the length of the fuselage was chosen. There was some consideration of a helicopter for greater lift and the ability to hover in mid-air. Due to the many variables that must be programmed for helicopter stability, it was decided to stay with an airplane design for ease of programming and reduced cost.

We have choose Turboprop engine for our design, we also like to have solar panel on top surface of the wing for system electrical backup.

IX. References

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